

Mohamed Elsayed Alshamy

Computer Vision & Robotics Engineer

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Objective

Computer Vision & Robotics Engineer with 3+ years of experience in **AI-powered autonomous systems**. Specialized in **real-time vision pipelines** (YOLO, OpenCV, TensorFlow) and **embedded AI deployment** on NVIDIA Jetson and Raspberry Pi. Strong background in **sensor fusion, ROS, and SLAM**, with a proven record of **leading teams** and winning **national and international robotics competitions**.

Education Qualification

Bachelor's Degree, Artificial Intelligence

Egyptian Russian University | ERU

Oct 2022 - Jun 2026

GPA: 3.15 / 4.0 | Grade: Very Good

Activities: ICPC - International Collegiate Programming Contest

Work Experience

Team Leader – NEURONIX Intelligence Team | ERU

Oct 2023 – Present

- Implementing AI-powered self-driving robotic vehicles and autonomous wheelchairs, achieving 92% lane detection accuracy and 35% fewer obstacle avoidance errors through YOLO + OpenCV pipelines and ROS-based sensor fusion (LiDAR, IMU, cameras).
 - Built a CARLA simulation and trained CNNs with reinforcement learning, improving lane-following success rate to 88%, while managing a team of 5 and presenting at international conferences.
- Top Skills:** Computer Vision (YOLO, OpenCV), Deep Learning (CNNs, RL), Robotics (ROS, CARLA), Sensor Fusion, Leadership

AI Team Leader – Neutrino Team | ERU

Oct 2023 – Jun 2025

- Led development of an autonomous vehicle with lane detection, obstacle avoidance, and traffic sign recognition, deploying YOLO-based models with multi-sensor fusion (camera + LiDAR + IMU) on NVIDIA Jetson.
 - Improved detection speed by **30%** and real-time navigation reliability by **25%**, achieving **1st place** in the Valeo Egypt Autonomous Vehicle Competition (2024).
- Top Skills:** Autonomous Driving, YOLO, Sensor Fusion, Embedded AI (NVIDIA Jetson), ROS

Team Leader – Hydrotron ROV Team | ERU

Oct 2022 – Jun 2025

- Directed the development of autonomous underwater ROVs with AI-powered perception, integrating sonar, LiDAR, and camera data for sensor fusion.
 - Built TensorFlow + OpenCV pipelines on NVIDIA Jetson, boosting underwater detection accuracy by **88%**, and led the team to **8th (2023), 6th (2024), and 4th (2025)** place in the MATE ROV International Competition.
- Top Skills:** Underwater Robotics, Computer Vision, TensorFlow, OpenCV, Embedded AI, Sensor Fusion, Leadership
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Internships and Certifications

- **Deep Learning – Computer Vision**, NTI (July 2025 – September 2025) – CNNs, YOLO, Mask R-CNN, Transfer Learning, TensorFlow, PyTorch.
 - **Embedded Systems**, ITI (July 2025 – August 2025) – C Programming, RTOS, Microcontrollers, Automotive Protocols.
 - **Artificial Intelligence Level 3**, Orange Digital Center (September 2024 – October 2024) – CV, NLP, GANs, VAEs, MLOps.
 - **HCIA-AI Certification**, Huawei ICT Academy (July 2024 – September 2024) – ML, DL, Model Deployment, CV Applications.
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Skills

- **Technical Skills**
 - Autonomous Systems Design: Intelligent robotic platforms for real-time interaction.
 - Real-Time AI Deployment: Optimizing ML models for embedded and edge devices.
 - Computer Vision: Object detection, tracking, image processing, scene understanding.
 - Sensor Fusion: Integration of LIDAR, sonar, IMU, GPS and cameras for navigation.
 - Deep Learning: CNNs, RNNs, GANs, reinforcement learning, motion prediction.
 - **Software & Tools**
 - **Programming:** Python, C, C++, Java, MATLAB
 - **Database Management:** SQL, NoSQL, MySQL
 - **Software Development:** OOP, SDLC, Docker
 - **AI Frameworks:** TensorFlow, PyTorch, Keras, OpenCV, YOLO
 - **Robotics:** ROS, NVIDIA Jetson, Raspberry Pi, Arduino, CUDA
 - **Tools:** Git, GitHub, Jupyter, Google Colab, Visual Studio Code, JetBrains IDEs
 - **Cloud & OS:** AWS, Google Cloud, Microsoft Azure, Linux, Windows
 - **Soft Skills**
 - Team Leadership, Problem Solving, Critical Thinking
 - Time Management, Public Speaking, Self-Motivation
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Languages

- Arabic: Native
- English: Fluent
- Russian: Intermediate
- French: Intermediate

